

Design and function

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General structure of the machine

Base frame	The machine is mounted on a base frame made of rustproof stainless steel with vibration-damping and variable-height supports.
Panelling	The front of the machine is equipped with protective doors with protective switches; the rear of the machine is equipped with a stainless steel housing with turnbuckles in the drive and installation areas. For further information on the safety equipment, please refer to the Chapter "Safety".
Production, drive and installation areas	The production area is separated from the drive and installation areas by means of a plate. This means that the machine's electric and pneumatic equipment is housed in the rear or lower part of the machine, depending on the machine type.

Limitations of the machine



Attention!

If hazardous, flammable/inflammable materials are processed, the machine owner is responsible for:

- safe processing,
- an extraction device in the working area.

Fundamentals of the machine design

Principle The machine/system is constructed according to the state of technology and the recognised safety rules. Nevertheless, when used, dangers may arise for the life and limb of the user or third parties, or adverse effects may occur for the machine and other property.

Using the machine The machine must only be operated in perfect technical working order, as well as intended, with attention given to safety and potential dangers under observation of the operating instructions! In particular, immediately rectify (or have rectified) malfunctions which could affect safety!

Intended use

Application of the machine The machine is designed for the punching of catheters.

The machine may only be operated in accordance with the intended use and with the products and/or formats specified in the associated work order. Only personnel qualified for the planned work at the machine may be used (see sections on operating personnel/user groups).

Any other use of the machine is considered a non-intended use. The manufacturer accepts no liability for damages which result from non-intended use; the user carries the associated risk.

The intended use also includes the adherence to the operating, maintenance and repair conditions specified by the manufacturer.

Spacial limits

In addition to the actual space required by the machine itself, there must be a clearance of min. 1.5 m all around the machine for opening of operating and maintenance access.

Time limits

The machine is designed for a total operating time of 10 years, or 20,000 operating hours, depending on which of these time limits occurs first.

Miscellaneous limitations

- The machine is to be operated in indoor facilities where the ambient temperature (at 50% relative air humidity) does not exceed 35°C and is not below 10°C. For more detailed specification of the ambient conditions, see Chapter 3 „Machine data“.
- The air humidity in the area surrounding the machine must only be between 20% and 80%.
- The machine is not suitable for operation in potentially explosive surroundings.
- The load-bearing capacity of the floor at the installation site of the machine must be min. 750 kg/m².

Residual risk

In spite of all precautions, the following residual risk remains:



Attention!

Risk of injury.

Protective doors are bridged when in Setup mode. As a result, it is possible to perform machine movements while the protective doors are open.

Only authorised persons, who are familiar with the machine operation and are aware of the possible dangers, may bridge protective doors.

- In Setup mode, the machine must only be operated by one person.
- Secure the machine against accidental operation by another person, e.g., affix a warning sign.



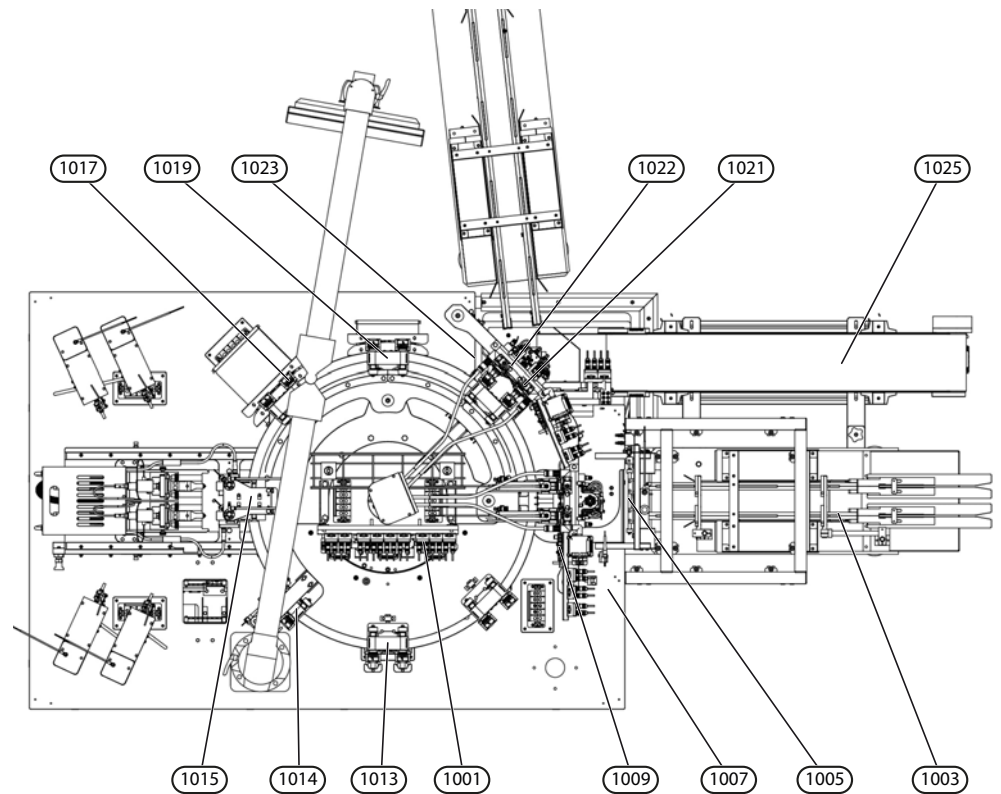
Warning!

Risk of crushing.

Tools can crush hands and fingers.

Do not reach in between the station tools.

Machine overview



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The machine consists of the following stations:

Station 1001 - Rotary Turret

Station 1003 - Catheter Feeding

Station 1005 - Catheter Transfer in

Station 1007 - Catheter Verification

Station 1009 - Catheter load onto mandrel

Station 1013 - Check straightness of Mandrels

Station 1014 - Control sitze manderls

Station 1015 - Eye Punching

Station 1017/1019 - Vision system front/rear

Station 1021 - Catheter unload from Mandrel

Station 1022 - Catheter Transfer out

Station 1023 - Reject station

Station 1025 - Outfeed

Functional description for regular operation

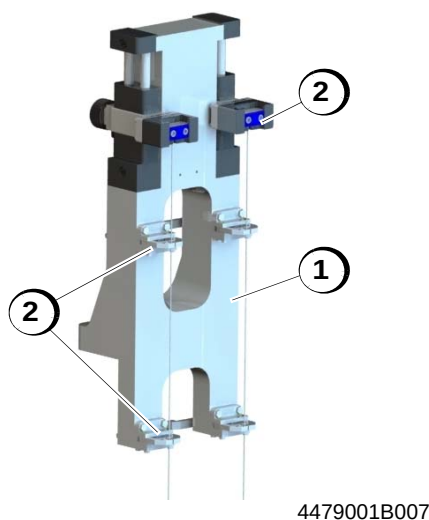
Station 1001 - Rotary Turret

A turntable (1) is mounted on the machine frame. The turntable is the central drive unit of the system. The turntable is driven by a servo motor.

Mounted on turntable (1) are eight holding towers (2). Each of the holding towers has room for two mandrels. Mandrels and catheters are moved through the machine with these holding towers. Grippers fix the catheters in position at the funnel.



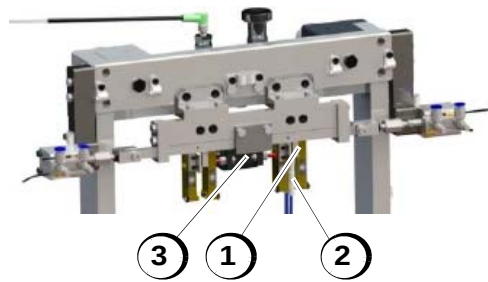
- Turntable (1)



- Holding tower (1)
- Mandrel (2)
- Gripper (3)

Station 1003 - Catheter Feeding

The catheters are fed continuously in two lanes into the machine. However, the machine can only process the catheters in intermittent motion. A slide lock blocks the infeed of catheters until a gripper is ready to receive the catheter. A gripper with centering bolt separates the catheters. The slide lock moves to the side. One catheter slides into the gripper. The gripper closes. The slide lock moves back to its original position, thereby blocking the infeed of any further catheters. The centering bolts retract to allow the catheter through, then they block the infeed again.

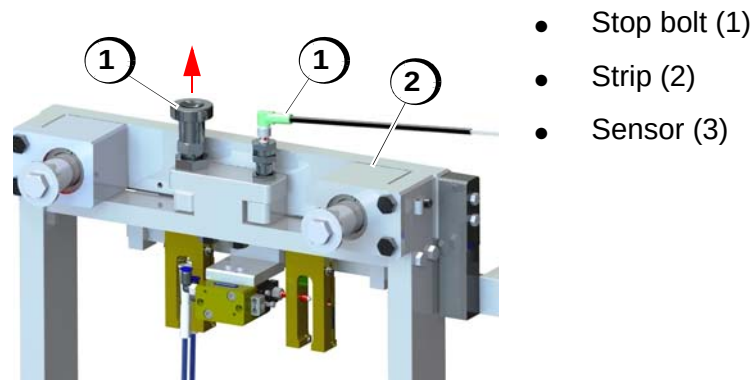


- Slide lock(1)
- Catheter (2)
- Gripper with centering bolt (3)

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Operation and adjustment

If the catheters are not accurately fed into the machine, a jam can occur. In order to facilitate clearing of this jam, it is possible to push the catheter separation unit apart.



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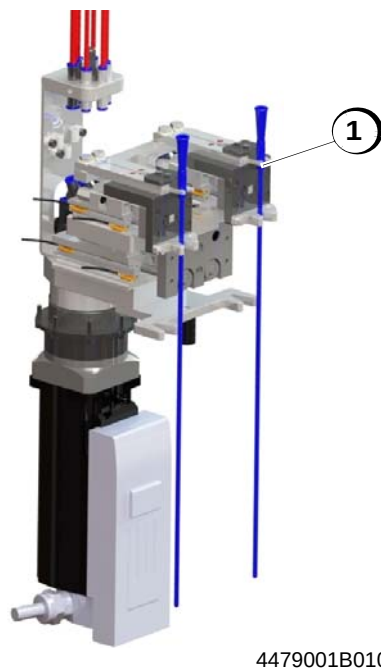
Pushing separation apart:

- ◆ Pull up stop bolt (1).
- The separation is unlocked.
- ◆ Push strip (2) with slide lock to the middle of the unit.
- ◆ Clear the jam.
- ◆ Push strip (2) with slide lock back to its original position.
- ◆ Allow stop bolt (1) to engage.
- ◆ Sensor (3) monitors the position of strip (2) with slide lock.

Station 1005 - Catheter Transfer in

Two grippers pick up one catheter each from the separation. A servo motor swivels the grippers by 90° into the checking station.

After checking has been completed, the grippers are swivelled by another 90° to the catheter insertion unit.



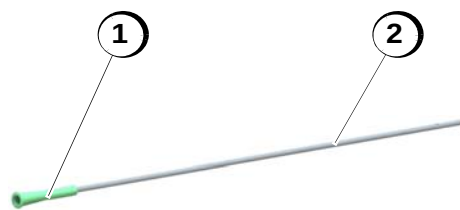
- Gripper (1)

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Station 1007 - Catheter Verification

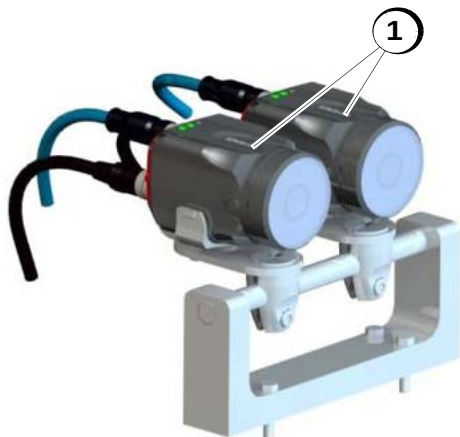
The „swivel catheter infeed“ unit positions the catheters in front of the cameras. The cameras detect the colour of the funnel.

If the colour is not correct, the machine stops. The incorrect catheter must then be manually removed.



Katheter

- Funnel (1)
- Pipe (2)



Kamera_Beschichtung

- Cameras (1) for colour detection



Observe!

For further information regarding operation, function and adjustment, please refer to manufacturer's operating instructions.

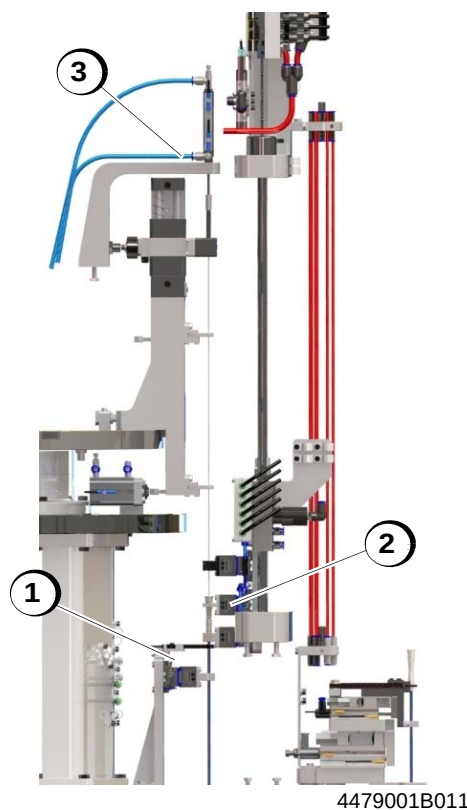
Station 1009 - Catheter loading onto mandrel

The catheter infeed swivels the catheters into position under the mandrels. Gripper (1) takes over the catheter.

The grippers (2) of the 'Pushing catheter onto mandrel' unit pick up the catheters in the bottom position and push them onto the mandrels during the subsequent upward motion.

In order not to damage the mandrels during this operation, a force limitation system (3) is installed above the mandrels. If the wire of the mandrel pushes onto a kink in the catheter during insertion, the wire is lifted. The force limitation system detects this and the machine is stopped.

The faulty catheter must be removed.



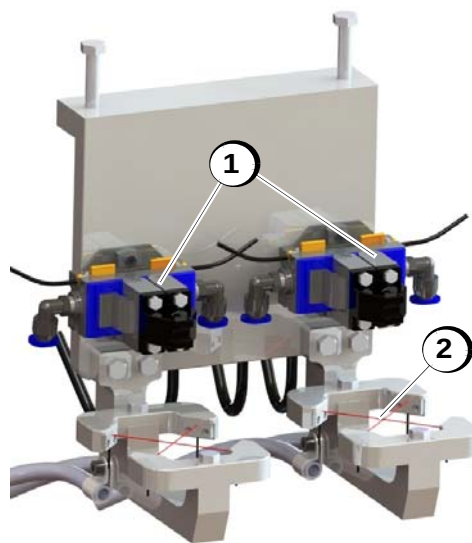
- Gripper (1)
- Gripper (2)
- Force limitation system (3)

Station 1013 - Check straightness of Mandrels

Catheters come in different lengths and diameters. If the diameter changes, matching mandrels must be inserted (see Chapter 10 „Conversion“). Each mandrel size has its own colour.

In the station, the mandrels are dipped. Grippers (1) fix the mandrels in position. The wire of the mandrel must now be accurately positioned in the crossing point of the fibre-optic light guides. If this is not the case, it cannot be guaranteed that the mandrel is straight. A different mandrel must be used.

When the straightness check has been successfully completed, grippers (1) open. The mandrels are moved to the next station, where a sensor checks the colour of the mandrel.



- Gripper (1)
- Crossing point (2)

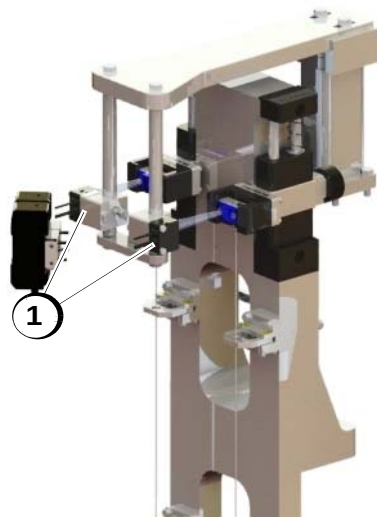
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Station 1014 - Check colour of Mandrels

Catheters come in different lengths and diameters. If the diameter changes, matching mandrels must be inserted (see Chapter 10 „Conversion“). Each mandrel size has its own colour.

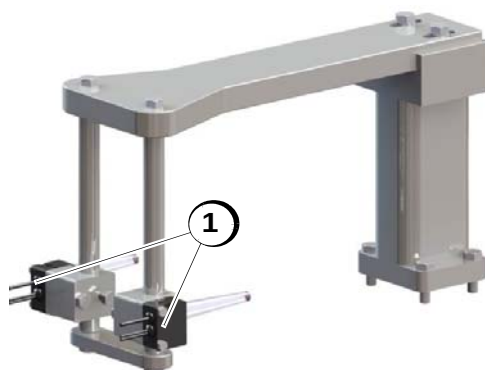
In Station 1014, the colour (and therefore the size) of the mandrels is checked.

If the colour is correct, the mandrels are transported to the next station. If the colour of the mandrel is not correct, the holding tower is indexed until it reaches Station 1013. At this station, the mandrels are checked for straightness.



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- Sensoren (1) prüfen die Farbe der Mandrels.



Mandrel_Farbe_2

- Sensoren (1) prüfen die Farbe der Mandrels.

Station 1015 - Eye Punching



Warning!

Warning of magnetic field.

Elektronic devices (cardiac pacemakers) could be disturbed or damaged.

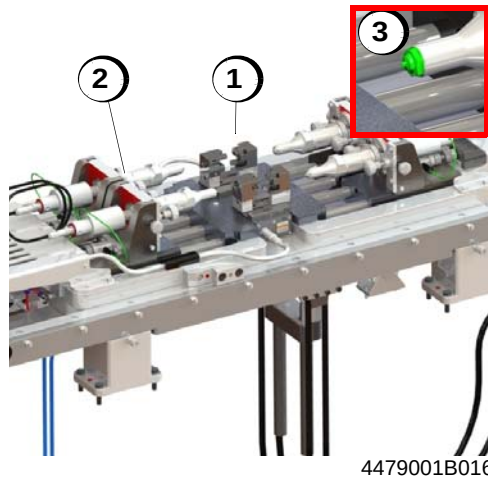
- Persons with cardiac pacemakers must keep away from the machine.
- Persons with cardiac pacemakers must not work at the machine.
- Sensitive electronic devices must be kept away from the LinMot (linear motor).
- Magnetic data carriers must be kept away from the LinMot.

Catheter and mandrel are swivelled with the holding tower into the processing position. One pair of parallel grippers (1) fix the catheter in position together with the mandrel.

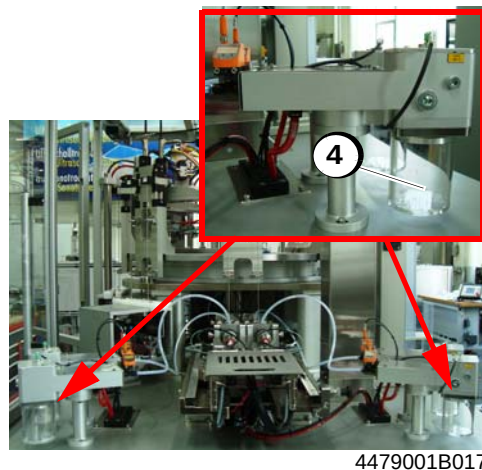
Four ultrasonic units (2) move into working position and punch two holes each into the catheter. The punching tools (3) are mounted at the ultrasonic units. These tools determine the size and shape of the holes.

A waste extraction unit is integrated in the ultrasonic unit to extract the waste from punching. The waste is fed out of the unit via hoses into a waste container (4).

Sensors, which are installed between unit and waste container, detect whether the waste passes on its way to the waste container. If this is not the case, this is registered by the control system and the corresponding catheter is automatically sorted out and rejected at the end of the machine.



- Parallel gripper (1)
- Ultrasonic units (2)
- Punching tool (3)



- Waste container (4)

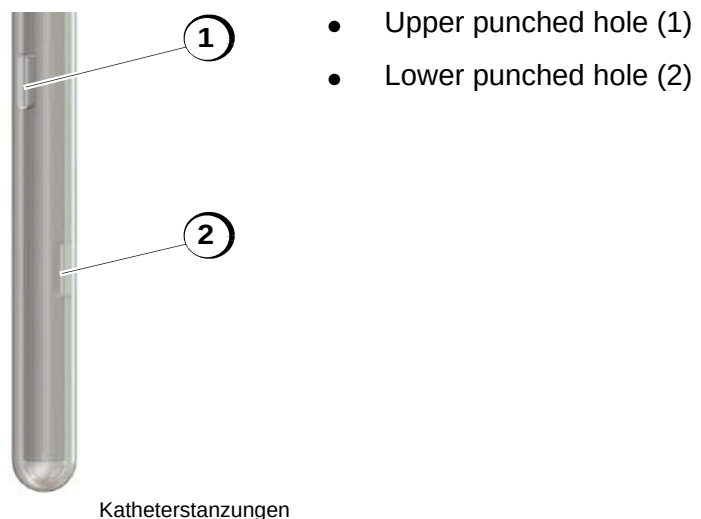
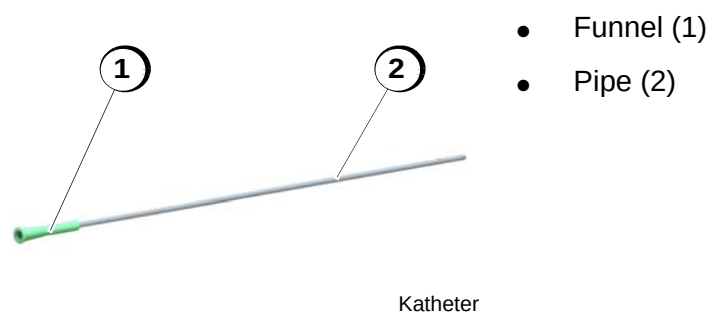
Station 1017/1019 - Vision system front/rear

In this station, cameras check the coating of the catheter and whether the holes have been punched correctly. In order to check the punched holes of the catheter, the mandrel is pulled out of the catheter a little just before reaching the station.

Each catheter must have two punched holes at different height.

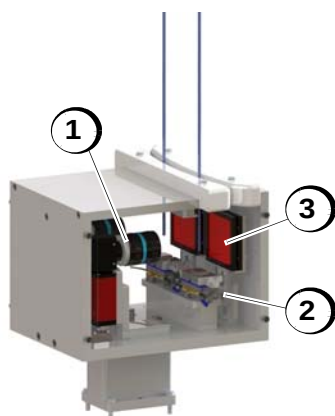
The lower hole is checked by a camera in Station 1017; the upper hole is checked by a camera in Station 1019.

In Stations 1021 and 1023 the catheters without fault are transferred for further processing, whilst the faulty catheters are sorted out and rejected.



Station 1017

Check of the lower punched hole

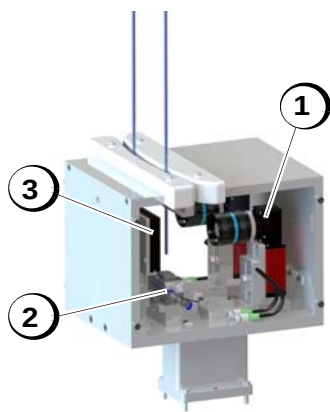


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- Camera (1)
- Gripper (2)
- Backlight (3)

Station 1019

Check of the upper punched hole



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- Camera (1)
- Gripper (2)
- Backlight (3)



Observe!

For further information regarding operation, function and adjustment, please refer to manufacturer's operating instructions.

Ceck coating of catheter pipe



- Camera (1) for detection of coating

Kamera_Farbe

**Observe!**

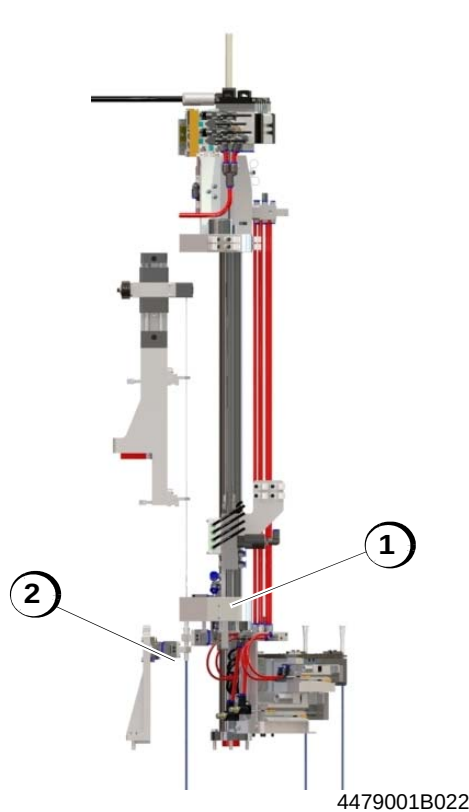
For further information regarding operation, function and adjustment, please refer to manufacturer's operating instructions.

Station 1021 - Catheter unload from Mandrel

At this station, the catheters are pulled off from the mandrels. The value for the stroke of the grippers (1) comes from a sensor of the catheter infeed station.

Grippers (2) take the catheters, whilst grippers (1) move back up.

The grippers for transfer to the sleever take the catheters and swivel them out of the machine. The catheters are transferred in accordance with the transfer option set in the program: They are transferred either to the sleever, to the transport belt, or into the waste box.

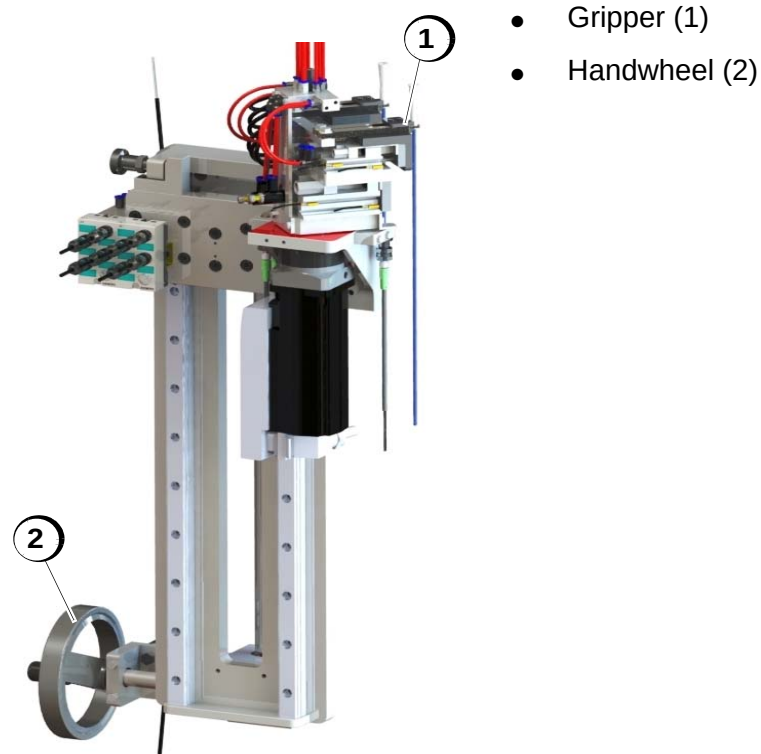


- Grippers (1)
- Grippers (2)

Station 1022 - Catheter Transfer out

Gippers (1) remove the catheters from the machine. Driven by a servo motor, they swivel the catheters to the required transfer position. This can be the transfer position to the sleeve, or to the transport conveyor. When the position is reached, the grippers transfer the catheters and then swivel back to the removal position.

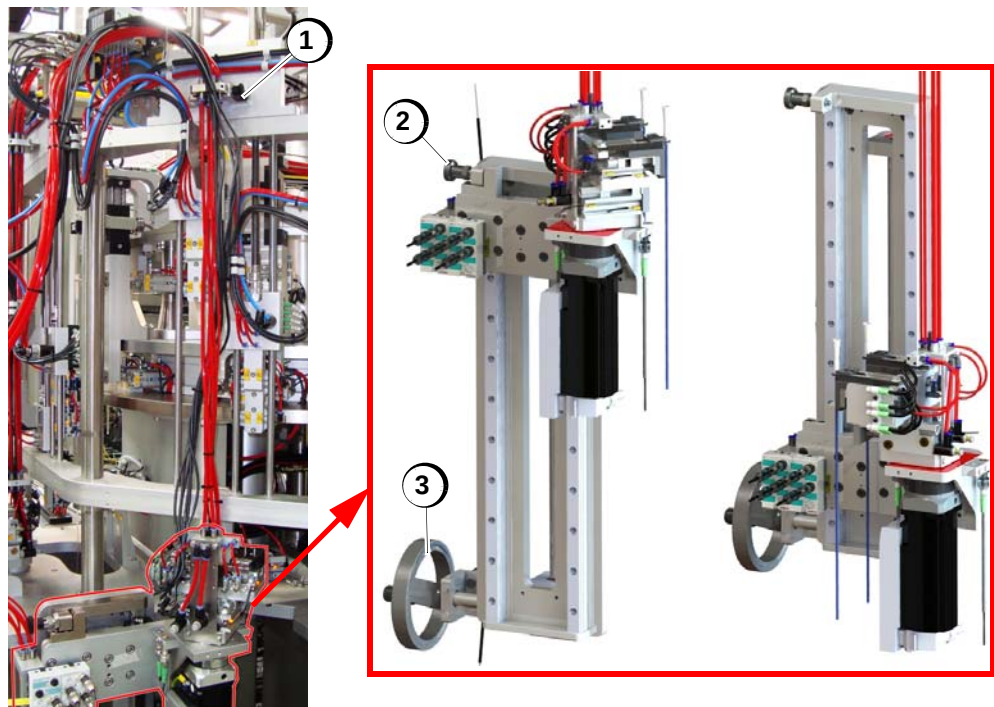
Optionally, another unit can be installed instead of the sleeve. The catheter removal unit must in this case be lowered by means of a handwheel (2).



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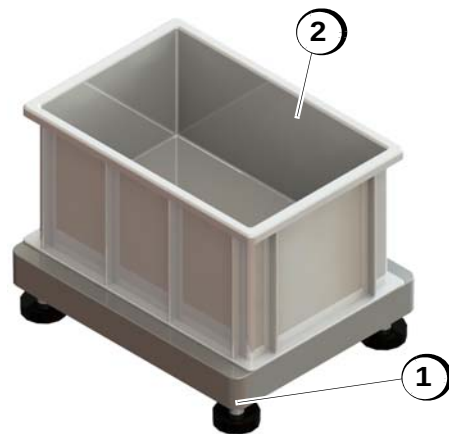
Lowering the catheter removal unit

- ◆ Pull out stop bolt (1).
 - ◆ Pull off supply lines with holder.
 - ◆ Pull out stop bolt (2).
 - ◆ Turn handwheel (3) in anti-clockwise direction.
- The catheter removal unit lowers.



Station 1023 - Reject

Weigher and stacking box are located underneath the 'catheter pull-off' unit. Any faulty catheters that arrive at the station are released by the grippers and drop into the stacking box directly underneath the station. The weigher determines the weight of the stacking box and transmits the data to the control system. When the stacking box is full, the weigher outputs a signal and the stacking box has to be exchanged or emptied.



- Weigher (1)
- Stacking box(2)

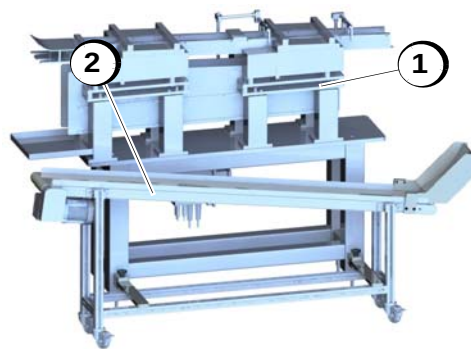
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Station 1025 - Outfeed

The finished catheters can be transported either to the downstream sleeve or onto a transport conveyor for discharged.

Grippers pick up the catheters from the machine, swivel them to the transport conveyor and release them there. They drop onto the transport conveyor.

The transport conveyor is mounted at the catheter feeder. To facilitate operation, it is possible to push the conveyor easily and quickly to one side.



- Catheter feeder (1)
- Transport conveyor (2)

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Dismantling of transport conveyor

- ◆ Pull of two bolts at handle.
- ◆ Open brakes at castors.
- ◆ Push transport belt to one side.
- ◆ Close brakes at castors.



Assembly of the transport conveyor is performed in reverse order.